

Matt McGlothlin

Honeywell Transmission Risk Air Monitor



Overview

Project name: HTRAM (Honeywell Transmission Risk Air Monitor)

Role: UX Lead

Timeline: 2021–2022 (or approximate timeframe)

Team: Cross-functional - hardware, firmware, analytics, BMS (building management) integration, business stakeholders

Mission: Design an intuitive user experience web and mobile application for a CO₂ / air-quality-based risk monitor that alerts classrooms and indoor spaces when ventilation conditions may increase airborne transmission risk of respiratory illness.

The TRAM measures CO₂, temperature, and humidity, uses three pre-programmed activity-level settings, and gives traffic light visual cues + sound alerts. It's a user-friendly monitor that alerts users when indoor air conditions may present increased risk factors for airborne virus transmission in schools, restaurants, etc. In a Honeywell "Success Story" document, the product was integrated with the Niagara framework to automate ventilation via BMS control based on alarm thresholds.

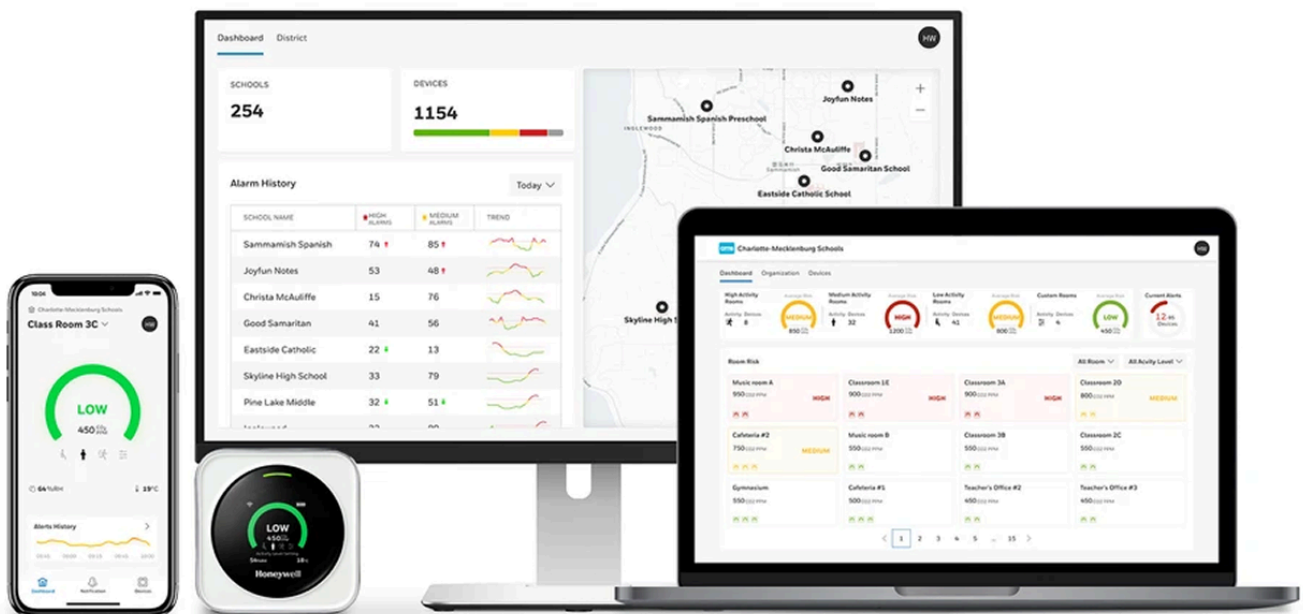
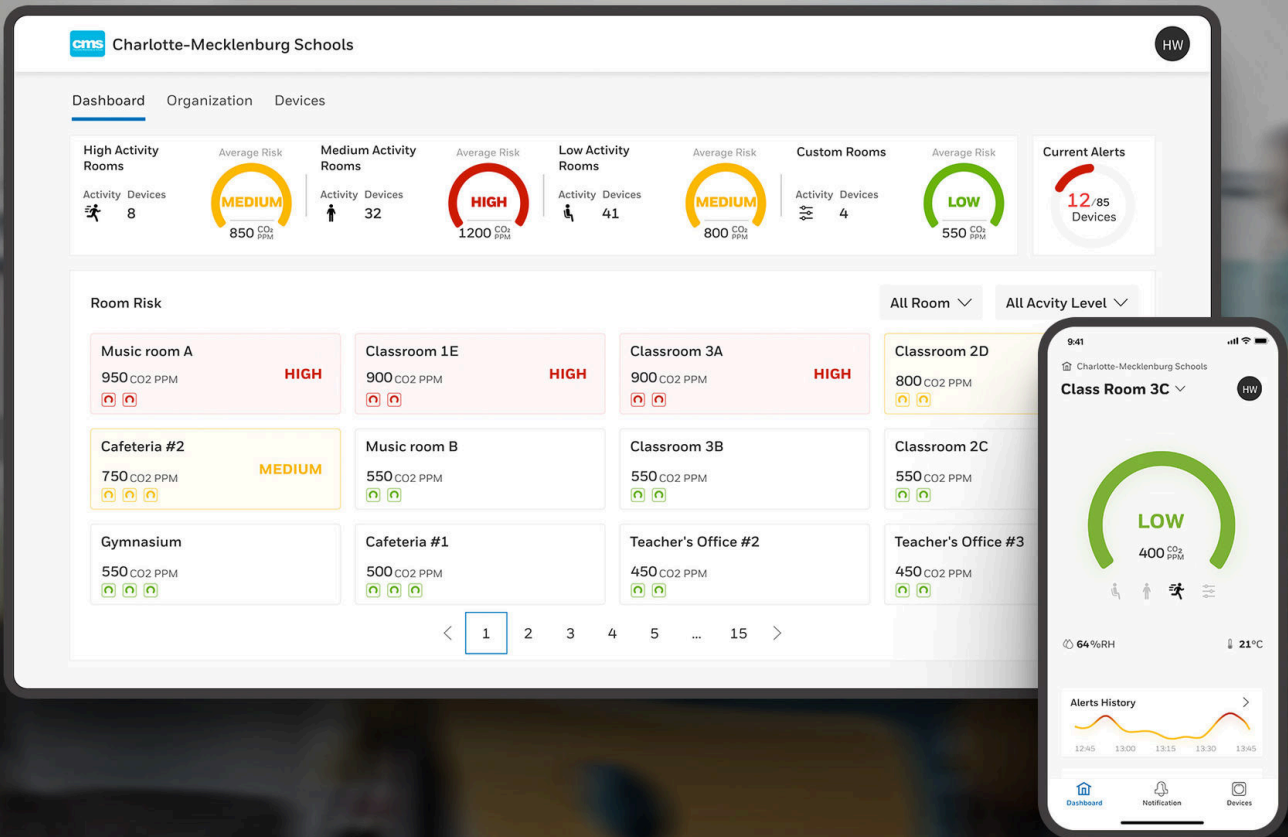
The project was organized into 4 core sprints for the initial MVO1. I worked with the global team to make sure developers core features made it into production. We were able to launch the application quickly as a result.

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- Lower barrier for non-technical users (teachers, facility managers) to understand, interact with
- Use visual + auditory cues (traffic light, beeps) to communicate alerts that are consistent with mobile and web applications
- Intuitively display average transmission risk scores at the district, school and room level
- Provide data visibility (dashboard, mobile app) for multi-device monitoring
- Enable integration with HVAC / BMS for automated mitigation



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indicators to know when ventilation was inadequate. Traditional HVAC systems are often run on schedules rather than continuously adaptive to real occupancy or CO₂ buildup.

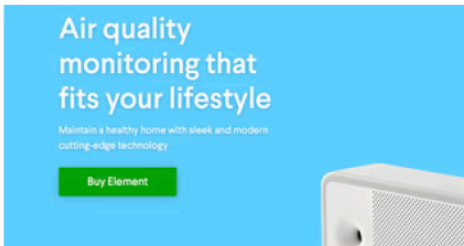
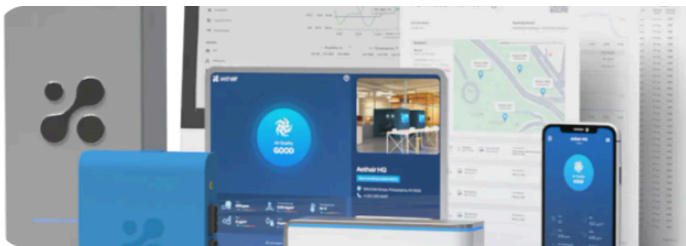
Challenges included:

- Non-technical users understanding on-device indicators
- Designing a simple and intuitive device setup via mobile and web app
- Scaling software to adapt seamlessly for large school districts, small schools and various room types
- Integrating with building systems (HVAC, BMS) for automated building response to CO₂ levels
- Creating data visualizations for users unfamiliar with technical metrics

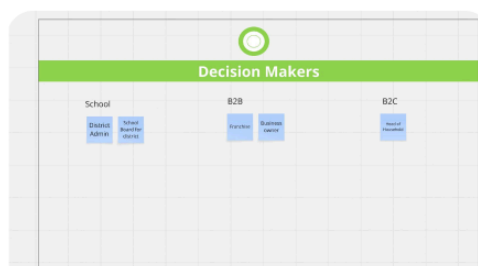
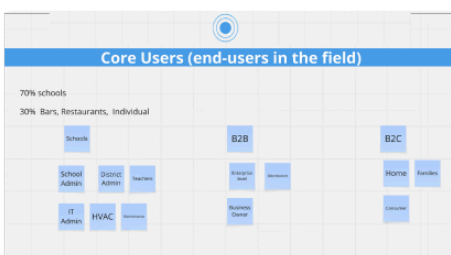
Research & Discovery

As UX Lead I helped conduct:

- **User & stakeholder interviews** with facility managers, teachers, school administrators — to understand their expectations, constraints, and workflows (preferences for visualizing and understanding data, how they would act on alerts with students in the room, setup the device with wifi and bluetooth).
- **Competitive / comparative analysis** by examining existing air-quality monitors, CO₂-based ventilation tools, and dashboards (e.g. Airthings, IQAir, other IAQ devices) and their UI/UX — to understand how effectively people interpret alerts, color coding, dashboards, data vs. simplicity tradeoffs.
- **Sensor & data domain research** by working closely with hardware / firmware teams to understand sensor accuracy, alarm thresholds, algorithms for transmission risk and wifi network integration limitations.
- **Technical constraints & edge case identification** of offline scenarios, battery life, connectivity issues.
- **Heuristic & accessibility audit** to ensure visual alerts (traffic light colors) on device are supported with redundant cues (icons, text, beeps) on desktop and mobile app so color-blind or low-vision users are accommodated.



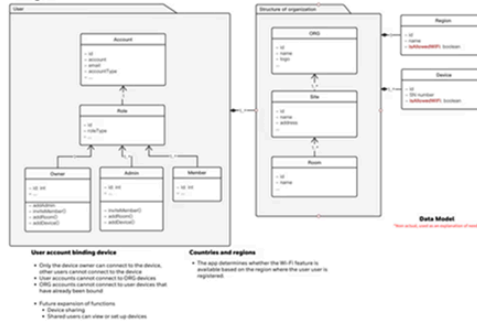
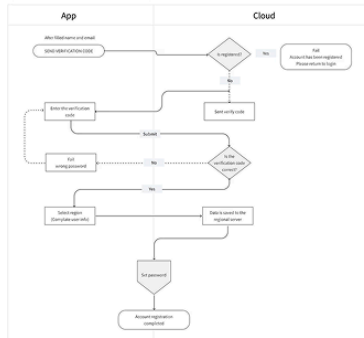
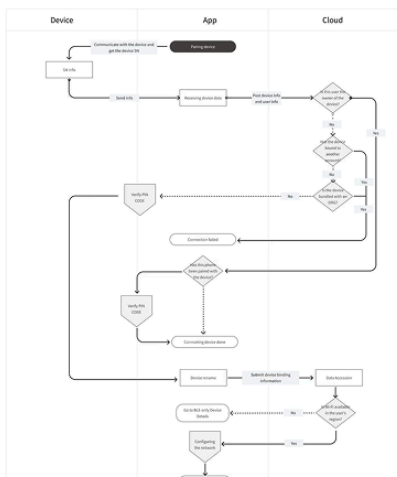
Product	Onboarding Experience	User Interface Design	Feedback & Alerts	Overall UX Takeaway
Airthings View Plus	Requires app pairing and dashboard registration	Clean but data-dense dashboard	Visual-only alerts	Excellent analytics, limited immediacy
uHoo Aura	Guided app setup, several permissions	Modern, data-rich interface	App notifications, no on-device cues	Great insights, but complex for schools



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During this phase I helped plan MVP sprints that included:

- **Workflows for adding a device by Bluetooth** for brand new devices without an owner, device access requests from users that don't have permission, device access requests from users that do have permission.
- **Workflow for Honeywell admin** enabling an admin to self-register a school and finally a workflow for an owner to share with another user.
- **Hifi designs** that for device screen UI, mobile app setup with BLE, web dashboard setup, and school/room/device setup screens. These took place during sprints 1-4.
- **Alerts that include visuals** both on device and on app for green / yellow / red states with icons and text consistency.
- **Onboarding / configuration flows** demonstrating how a user sets the activity mode, links the device to a network, assigns room names and activity type
- **Dashboard designs** for multi-device monitoring (across classrooms, schools, devices) with trend graphs, alarm logs, filtering device types and room types.
- **Prototypes for usability testing** as well as interactive flows for configuring a device, interpreting an alert, and taking action.

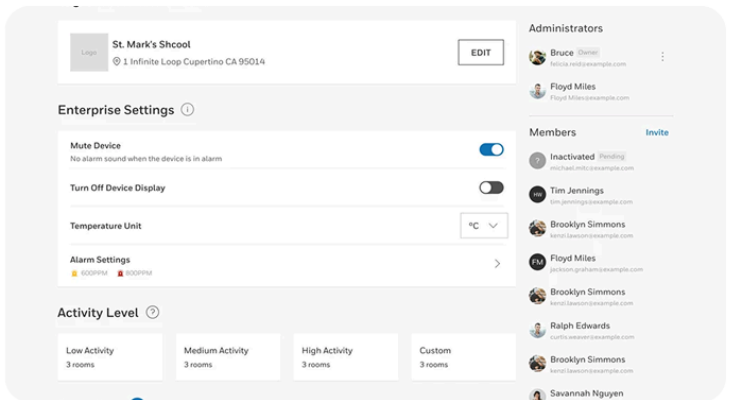
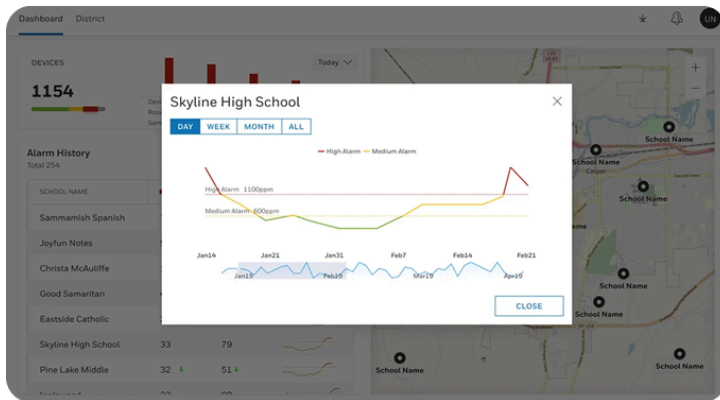
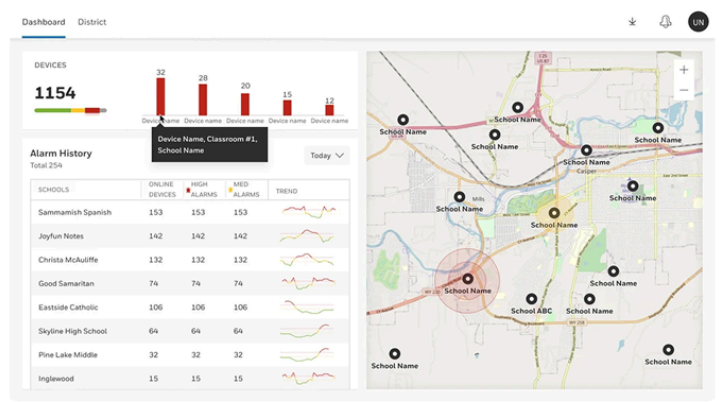
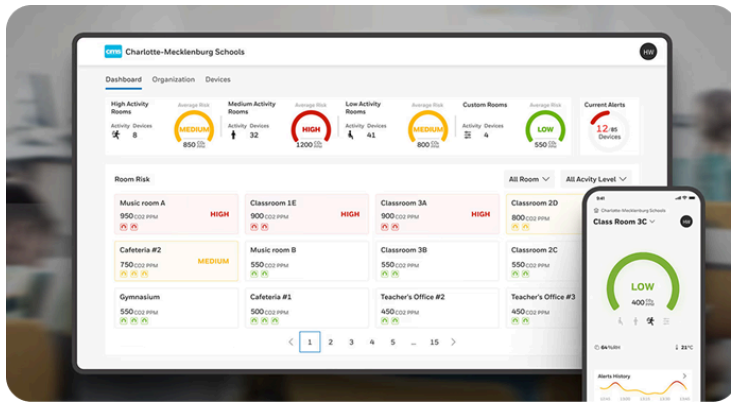


Validation & Testing

I conducted usability testing and validation with facility managers or education users by presenting them prototype flows (e.g. connecting device, reacting to red alert, configuring activity) that informed the following:

- **Alert comprehension trials** to test whether users understand what “red / yellow / green”, icons and CO2 concentration values mean as well as what action to take at each level.
- **Iterations that refined iconography**, data visualizations, terminology, settings and thresholds (within allowable hardware constraints).
- **Integration tests that** simulate alarms and confirm the user sees the expected change in dashboard/alerts.

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Design Delivery

I help plan MVO1 as I led the design of comprehensive solutions across device, mobile, and web platforms throughout sprints 1-4. We delivered:

Alert state systems — Consistent visual language for green/yellow/red states across device and app interfaces, including coordinated icons and messaging

High-fidelity interfaces — Device screen UI, mobile app with Bluetooth setup, web dashboard, and school/room/device configuration screens

Onboarding flows — Step-by-step user journeys for setting activity modes, linking devices to networks, and assigning room names and activity types

Multi-device dashboards — Comprehensive monitoring interfaces with trend graphs, alarm logs, and filtering capabilities across classrooms, schools, and device types'

Interactive prototypes — Testable flows for device configuration, alert interpretation, and response actions used in usability validation sessions

These deliverables established a cohesive design system that scaled from individual device interactions to district-wide monitoring while maintaining consistency and usability for non-technical users.

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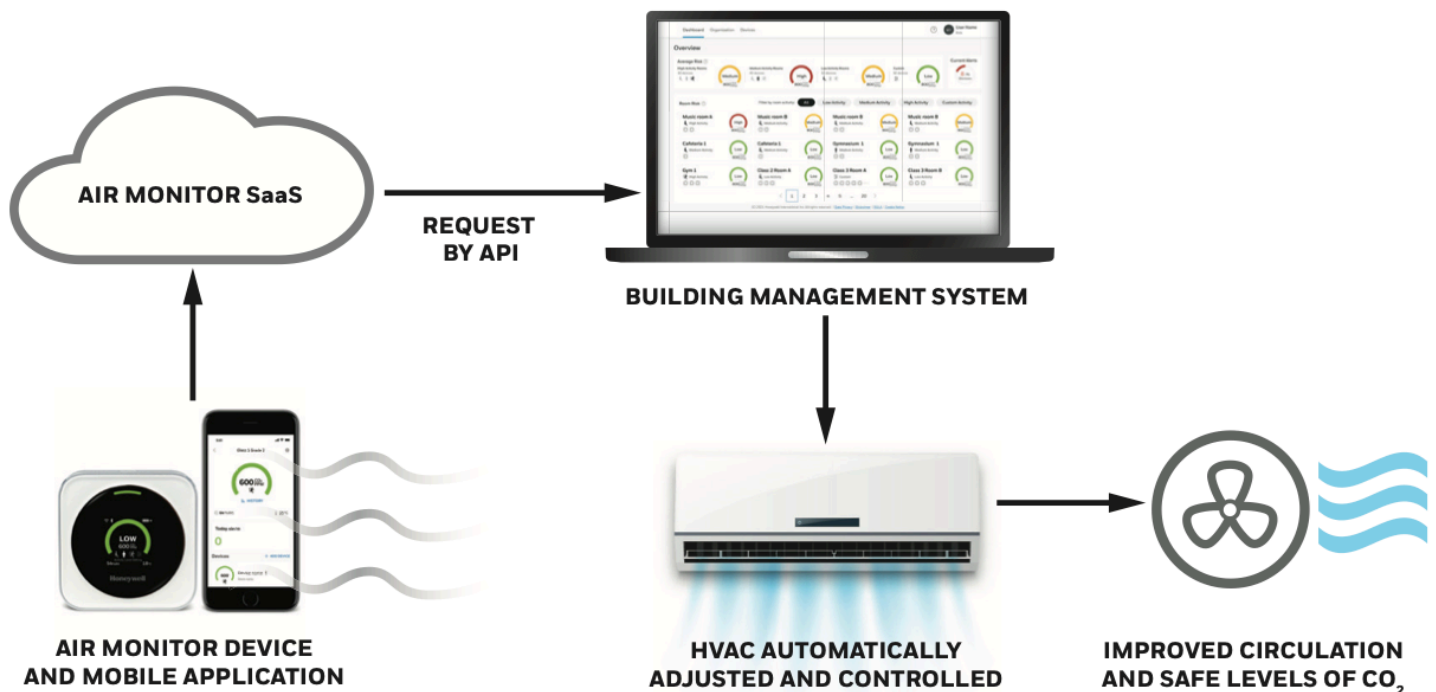
Outcome & Impact

The device shipped with a simple, widely adopted UI that non-expert users could understand and act upon.

Positive outcomes from my work included:

- **Increases in school administrator trust** in indoor air quality monitoring in school / public settings.
- **Facility team empowerment** to setup devices seamlessly with existing infrastructure.
- **Proactive teacher management** of airborne illness risk in the classroom without disturbing the classroom.
- **Air monitoring accuracy** for school administrators for a holistic depiction of transmission risk at the school and event district level.
- **Brand reinforcement** for broader “Healthy Buildings” messaging within Honeywell.

The project established foundational UX patterns for Honeywell air monitoring devices in educational environments, and the TRAM product was adopted in pilot school / building settings within Honeywell. The system's integration with Honeywell's BMS (e.g. Niagara) allowed automated ventilation adjustments — turning red alerts into active mitigation (this is documented publicly in a press release for the Charlotte Headquarters)



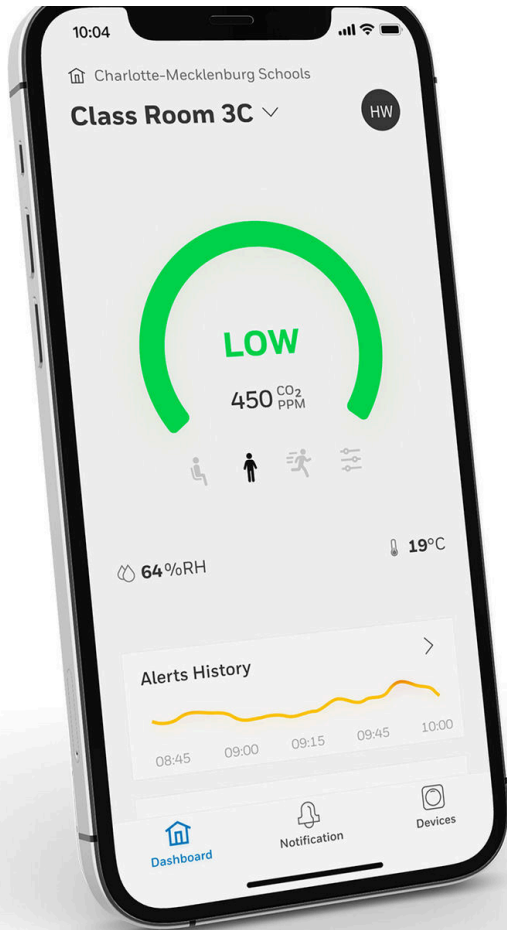
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Reflection & Learnings

Designing for non-technical audiences pushed me to simplify the ui and setup process. I learned the importance of consistent and redundant signaling (beeps, icons, text) in real-world alert systems both on device and on-app. Balancing data richness vs usability taught me that I can't expose too much sensor detail to overwhelm novice users. Collaboration across hardware, firmware, analytics, and building systems pushed me as a UX lead to be domain fluent. This experience reinforced designing for trust, clarity, and inclusive communication in safety- and health-software solutions with IoT devices.

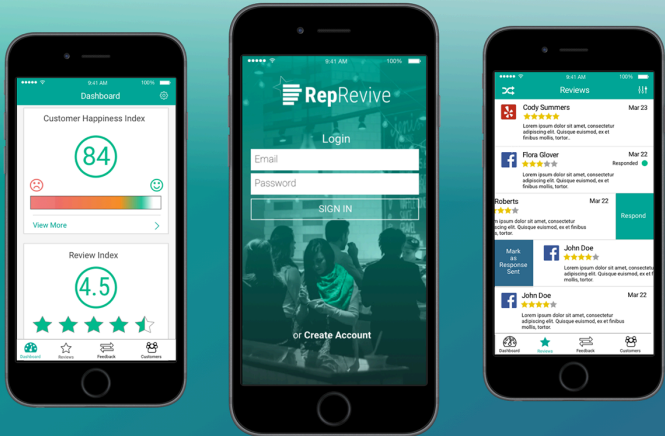
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